Results Overview

• Designs generated: 1000

• Designs passing all filters: 29

• MMCIF files of final designs are in:

/home/ayushregmibagale/gbm-protein-switch/boltzgen/output/sox9/final_ranked_designs/final_20_designs

• Metrics and sequences of 20 final designs are in:

/home/ayushregmibagale/gbm-protein-switch/boltzgen/output/sox9/final_ranked_designs/final_designs_metrics_20.csv

• Metrics of all designs are in:

/home/ayushregmibagale/gbm-protein-switch/boltzgen/output/sox9/final_ranked_designs/all_designs_metrics.csv

You can rerun filtering (very quick), using this command with changed parameters:

```
-- boltzgen run input_spec.yaml --steps filtering --config filtering budget=60 alpha=0.05
```

You can also rerun filtering in a jupyter notebook if you want using `filter.ipynb`

What was run to produce this in the Filter task:

1. Filtering: each design is evaluated against mandatory thresholds.
2. Ranking: for every metric we compute its rank, then scale it by the metric's inverse-importance weight.

Designs with fewer passed filters are automatically penalised because the ranking key is the pair (num_filters_passed, metric). The overall quality score is the worst (maximum) of these scaled ranks.

The 10 best designs form the Top set.

3. Diversity: a lazy-greedy algorithm selects 20 designs that jointly maximise quality and minimise sequence similarity (sequence-identity distance). The trade-off is controlled by α = 0.001:

• α = 0 → 100 % quality focus (same as Top set)

• α = 1 → 100 % diversity focus (ignores quality)

• Quality – composite of metrics such as iPTM, salt-bridges, ΔSASA, etc. Each metric has an "inverse importance" weight (see "Sorting Criteria" table). A larger weight divides the rank by a bigger number and therefore down-weights that metric.

• Diversity – 1 − sequence identity between designs.

We use α = 0.001, meaning 100 % emphasis on quality, 0 % on diversity.

CSV Column Reference

Column	Description
id	filename to retrieve the file
design_sequence	designed amino acids (may be subset of chain)
designed_chain_sequence	full sequence of the chain containing designed residues (recommended for synthesis)
num_design	number of designed residues
secondary_rank	intermediate rank from the sorting procedure
design_ptm	predicted TM score for intra-design contacts (higher = better)
design_iptm	predicted TM score for design–target contacts (higher = better)
design_to_target_iptm	same as design_iptm but for multi-chain designs
min_design_to_target_pae	minimum PAE between design & target (lower = better)
plip_saltbridge	number of salt-bridge interactions
plip_hbonds	number of hydrogen-bond interactions
plip_hydrophobic	number of hydrophobic interactions
delta_sasa_original	ΔSASA when binder present vs absent
delta_sasa_refolded	same as above but on the refolded structure

Filtering & Sorting Criteria

This section first lists threshold filters that every design must pass (Filtering Criteria table)then explains how designs are ranked (Sorting Criteria table) weighted by their inverse importance.

Filtering Criteria

feature	lower_is_better	threshold	Pass
has_x	True	0.0	1000
filter_rmsd	True	2.5	377
filter_rmsd_design	True	2.5	432
designfolding-filter_rmsd	True	2.5	178
ALA_fraction	True	0.3	973
GLY_fraction	True	0.2	1000
GLU_fraction	True	0.2	557
LEU_fraction	True	0.3	1000
VAL_fraction	True	0.2	1000

Sorting Criteria

Metric	Inverse Importance
design_to_target_iptm	1
design_ptm	1
neg_min_design_to_target_pae	1
plip_hbonds_refolded	2
plip_saltbridge_refolded	2
delta_sasa_refolded	2

Results Summary

Aggregate statistics

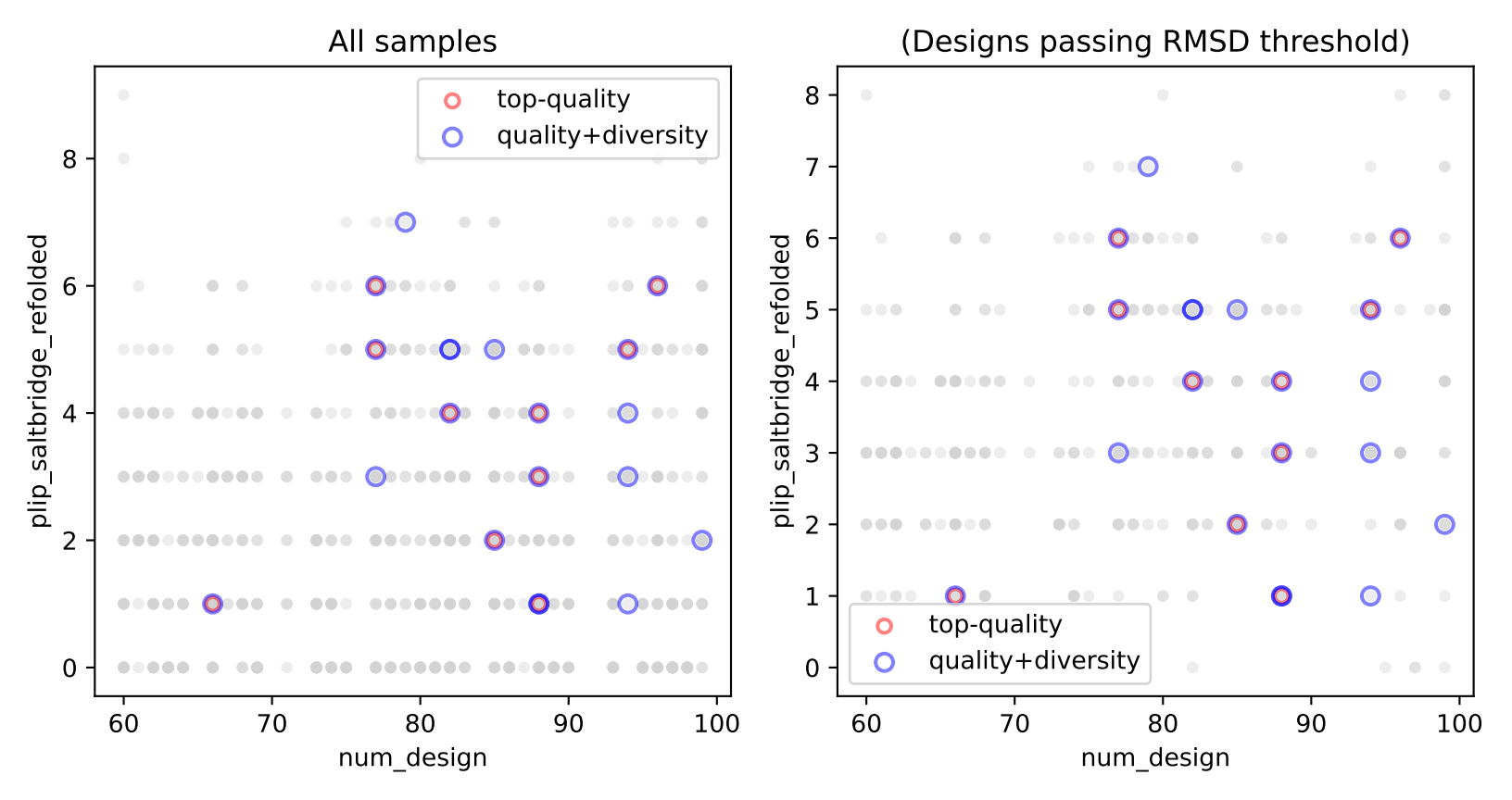
Quick numeric overview comparing (i) ALL incoming designs, (ii) the top-10 highest-quality designs, and (iii) the 20 quality+diversity designs produced by the lazy-greedy selection.

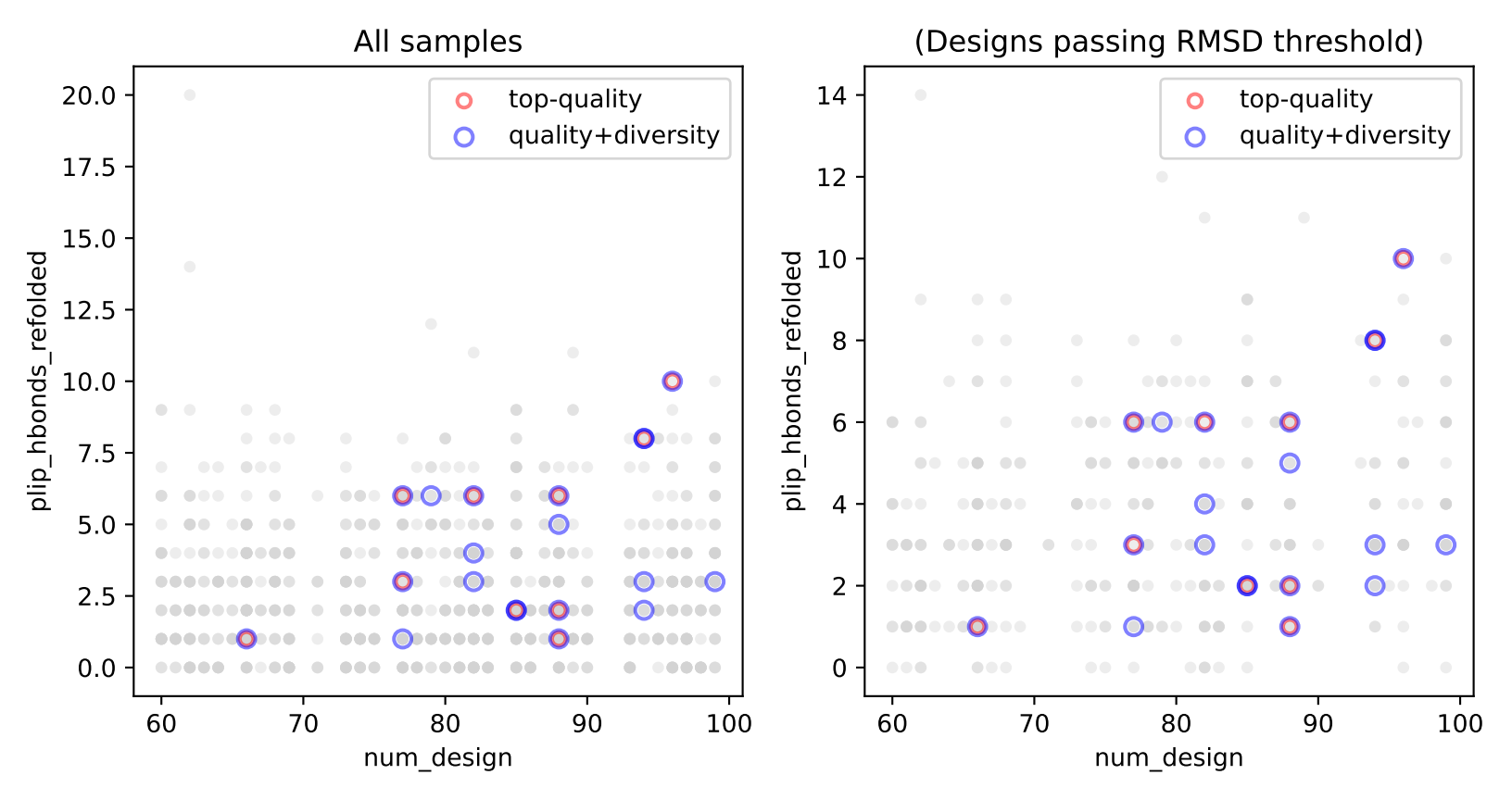
Metric	Mean	Mean top 10	Mean top 20 diverse
Num designs	1000	-	-
num_design	79.950	84.100	85.750
filter_rmsd	7.727	0.963	1.001
designfolding-filter_rmsd	5.935	1.959	1.779
design_iptm	0.428	0.658	0.626
design_to_target_iptm	0.428	0.658	0.626
min_design_to_target_pae	6.169	2.255	2.573
delta_sasa_refolded	947.863	1400.097	1376.053
plip_saltbridge_refolded	2.204	3.700	3.650
plip_hbonds_refolded	2.664	4.500	4.100

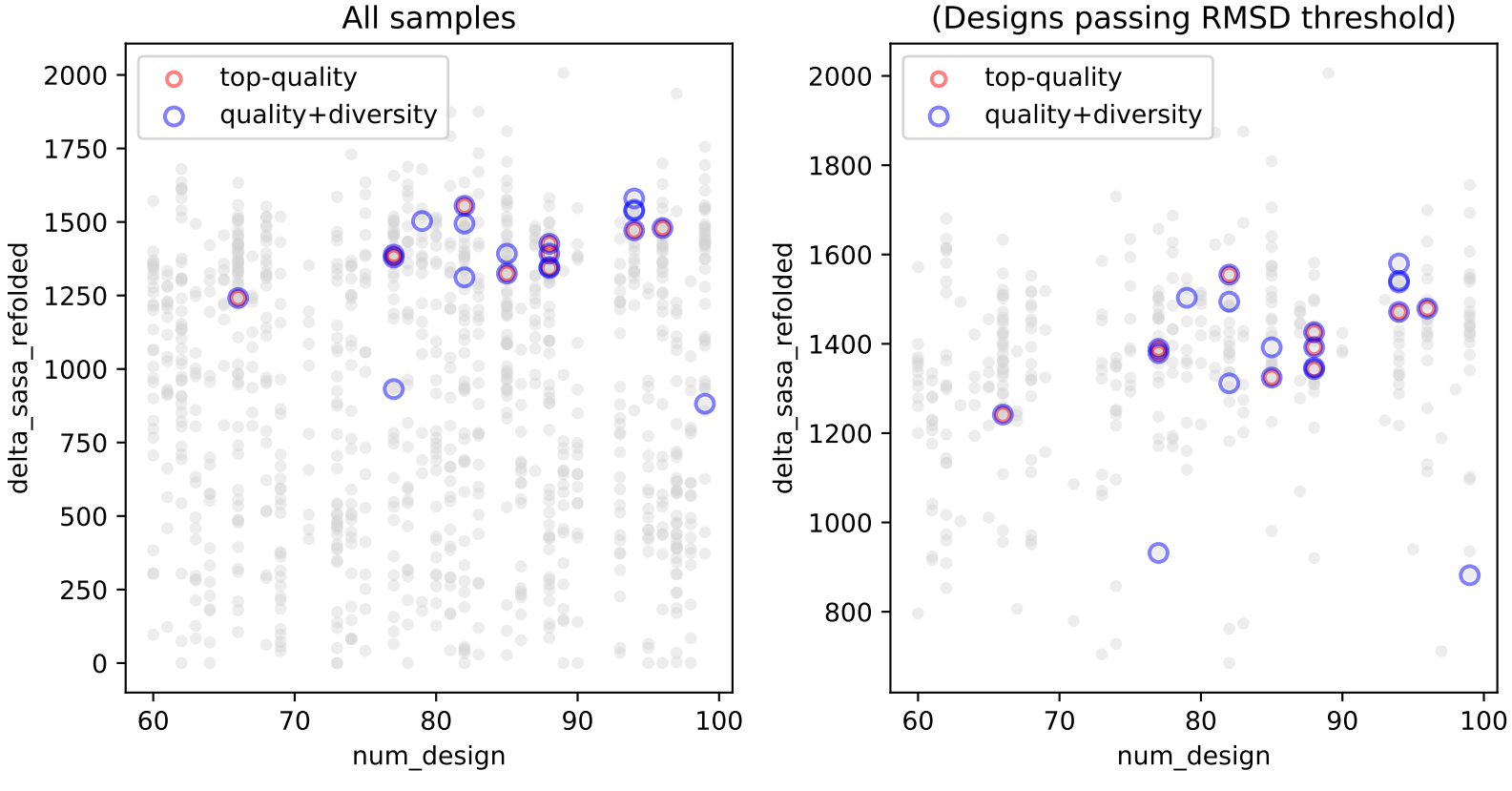
Scatter Plots

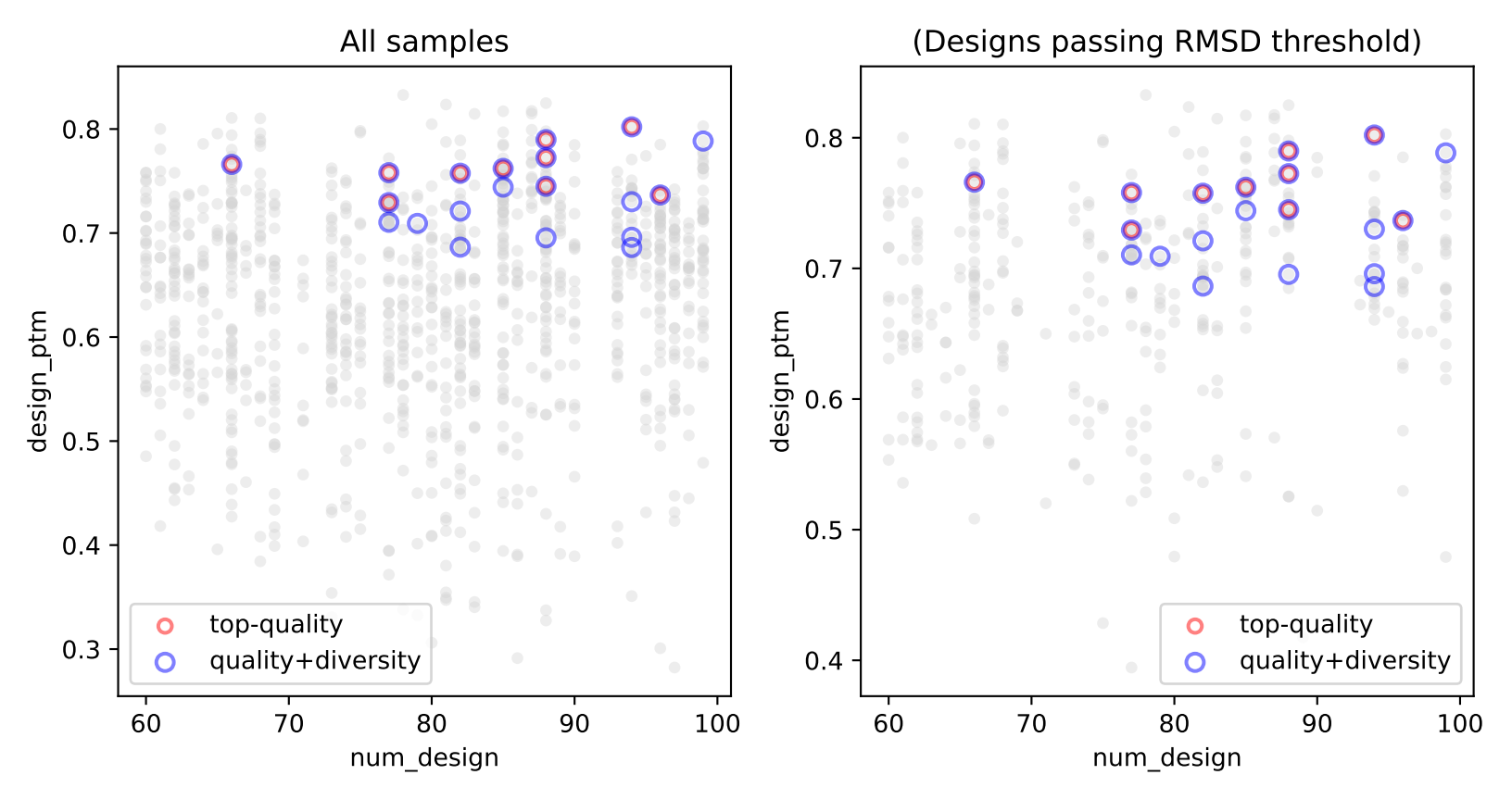
Metric relationships

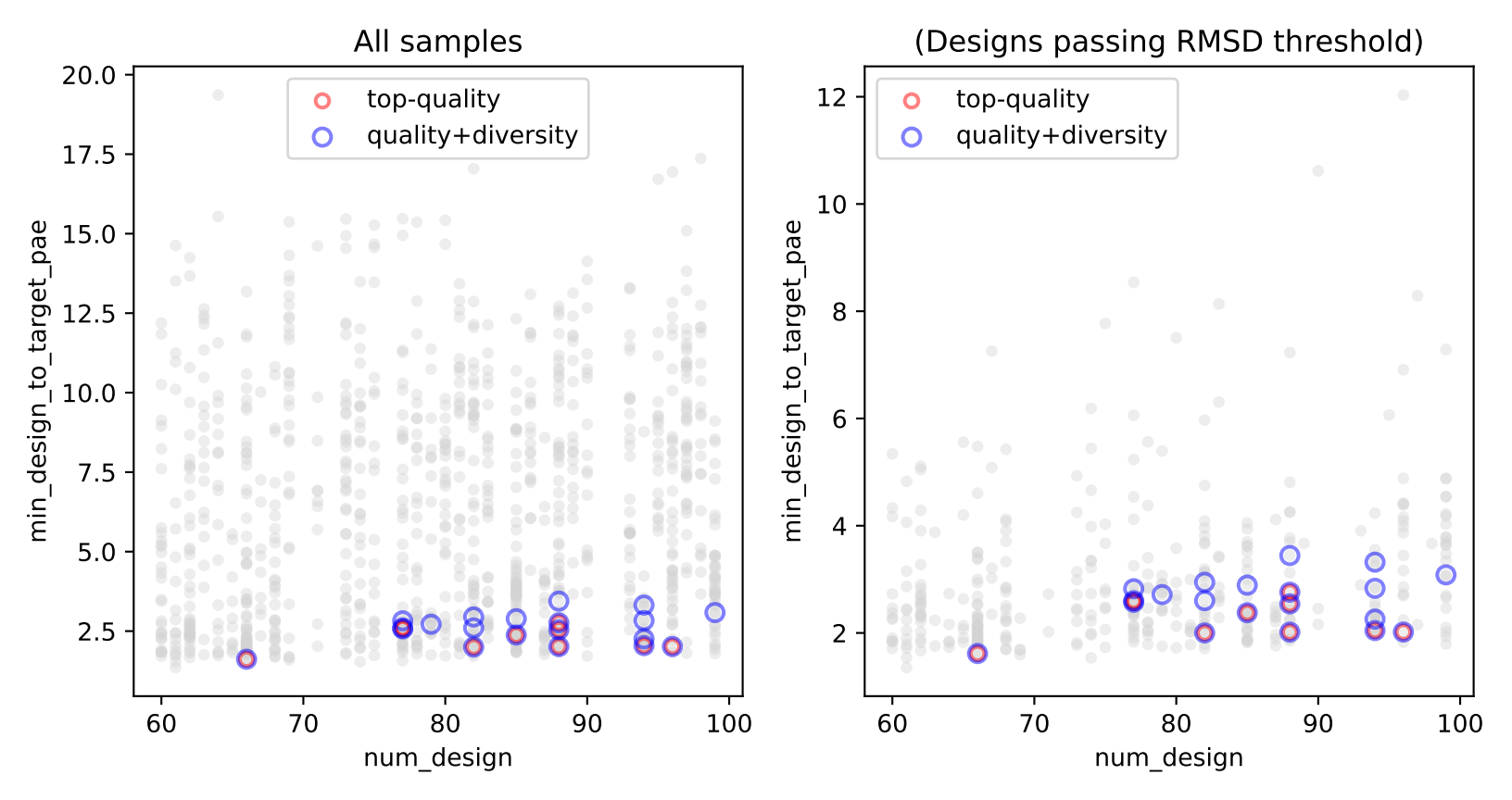
Each scatter page contains two panels:
• Left – all designs (grey) with overlays of Top (red) and Diverse (blue).
• Right – same but limited to designs passing the RMSD filter.

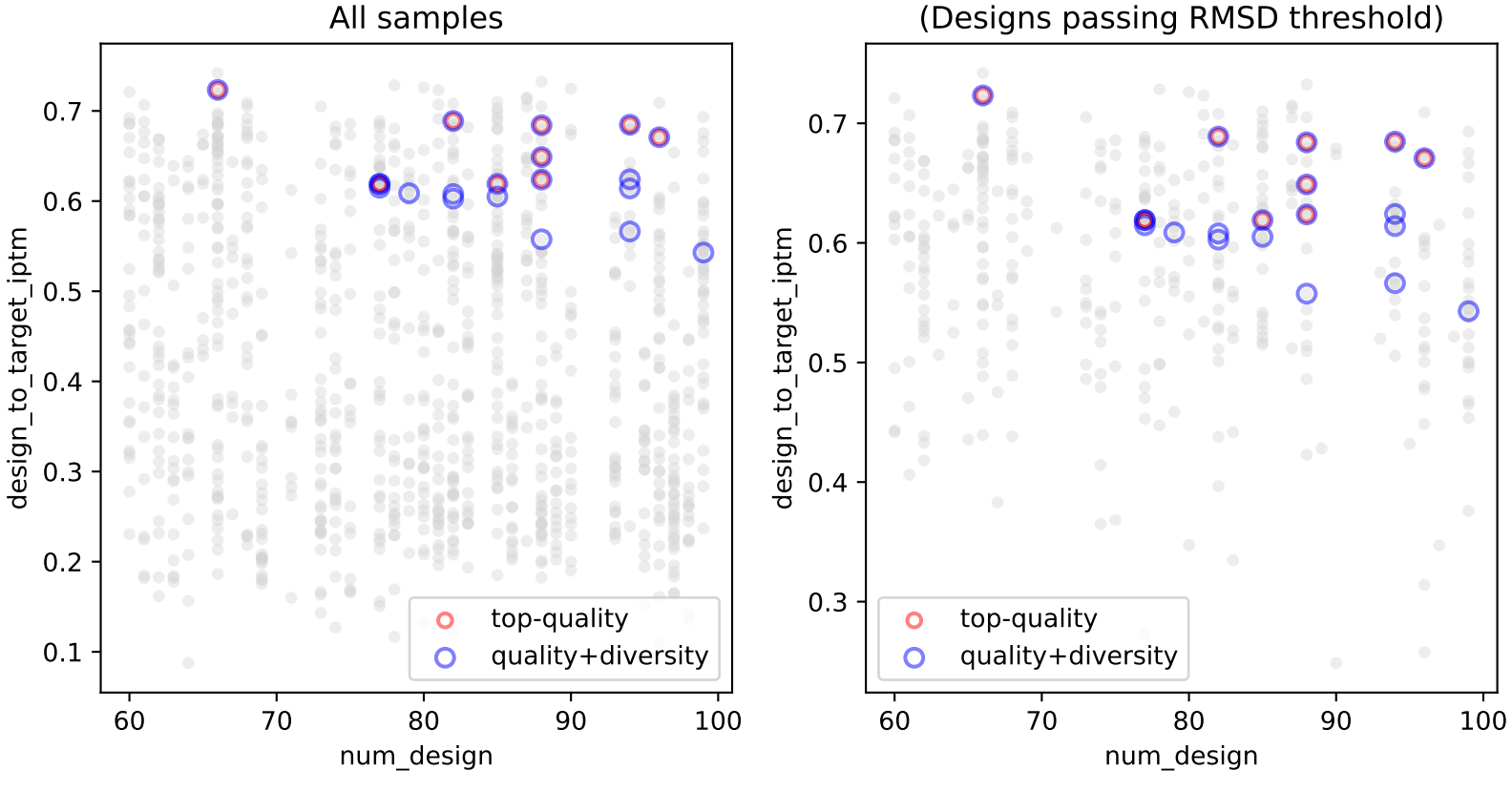

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
plip_saltbridge_refolded
num_design
0
2
4
6
8
60
70
80
90
100
1
3
5
7

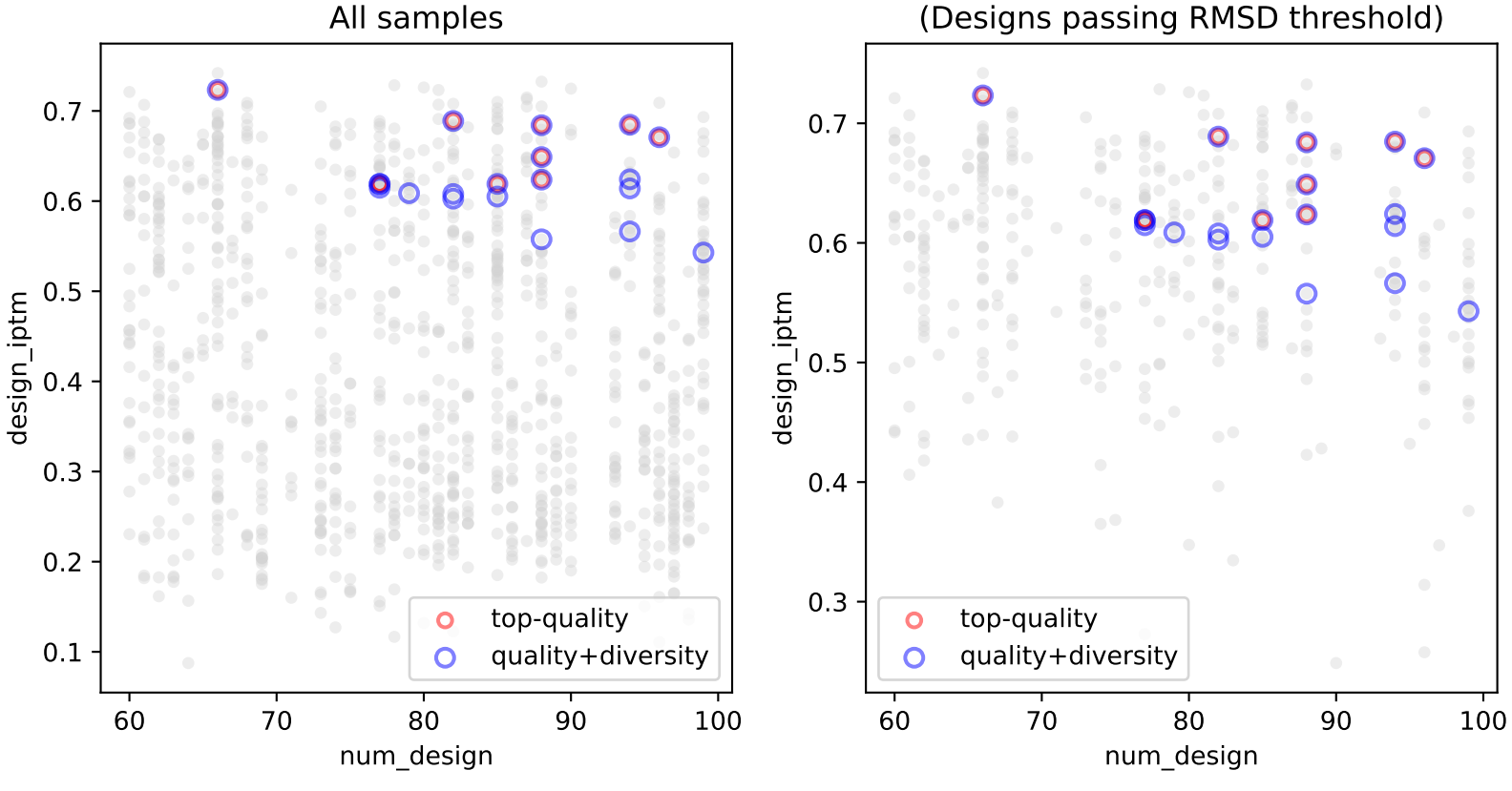

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
plip_hbonds_refolded
num_design
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
2
4
6
8
10
12
14
60
70
80
90
100

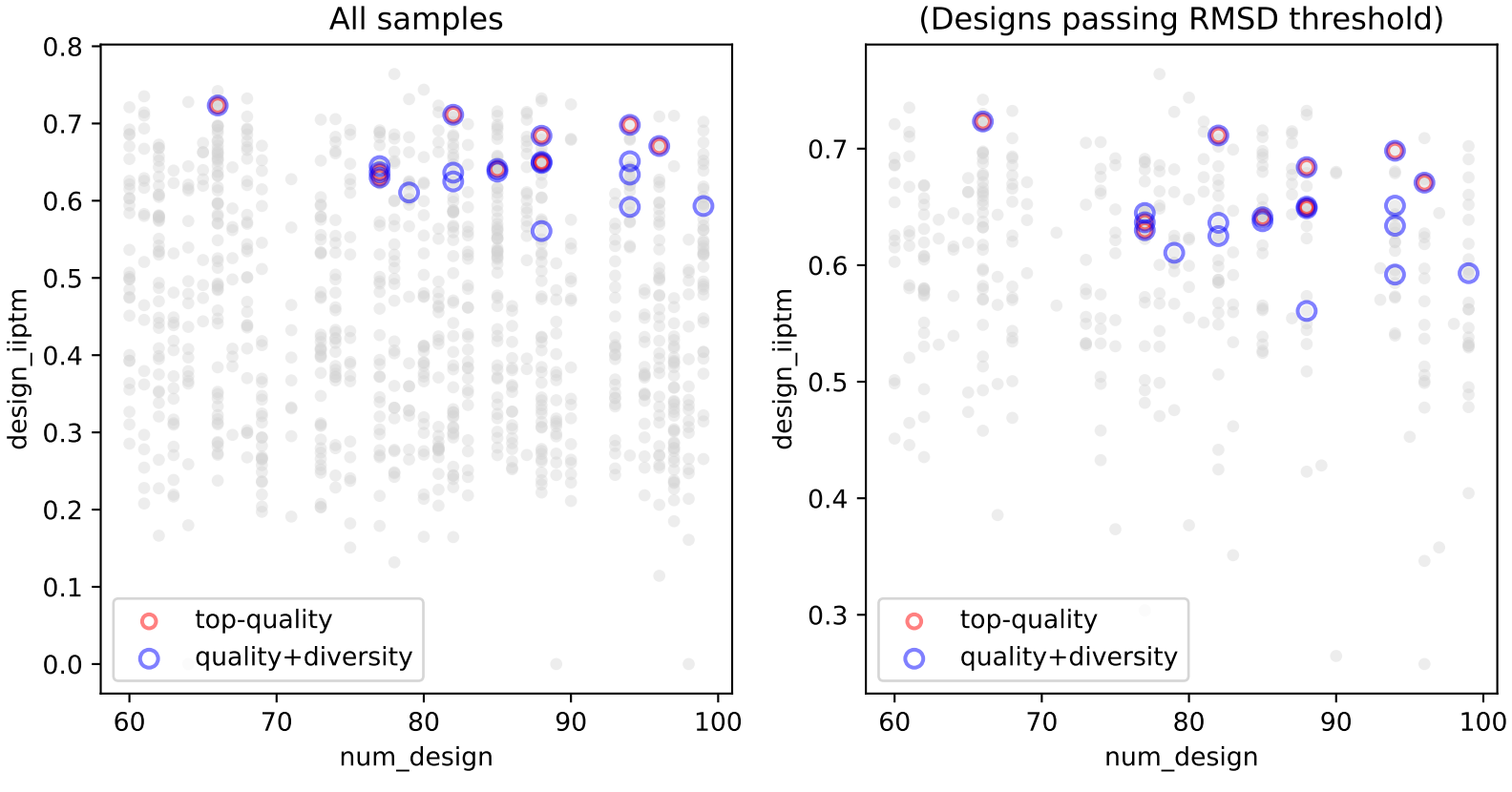

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
delta_sasa_refolded
num_design
2000
1750
1500
1250
1000
750
500
250
0
1800
1600
1400
1200
800
60
70
80
90
100

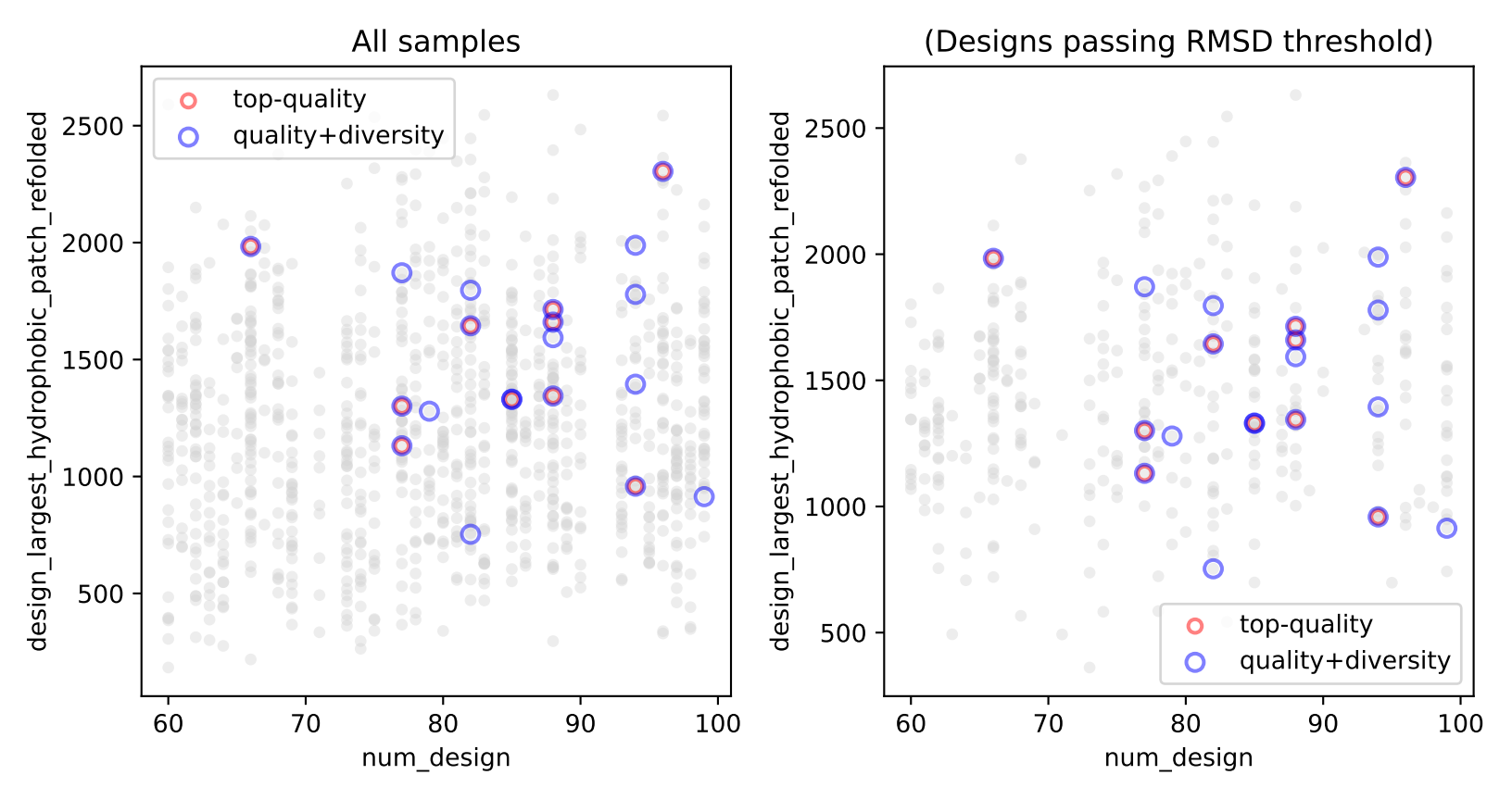

All samples
(Designs passing RMSD threshold)
design_ptm
num_design
top-quality
quality+diversity
0.8
0.7
0.6
0.5
0.4
0.3
60
70
80
90
100

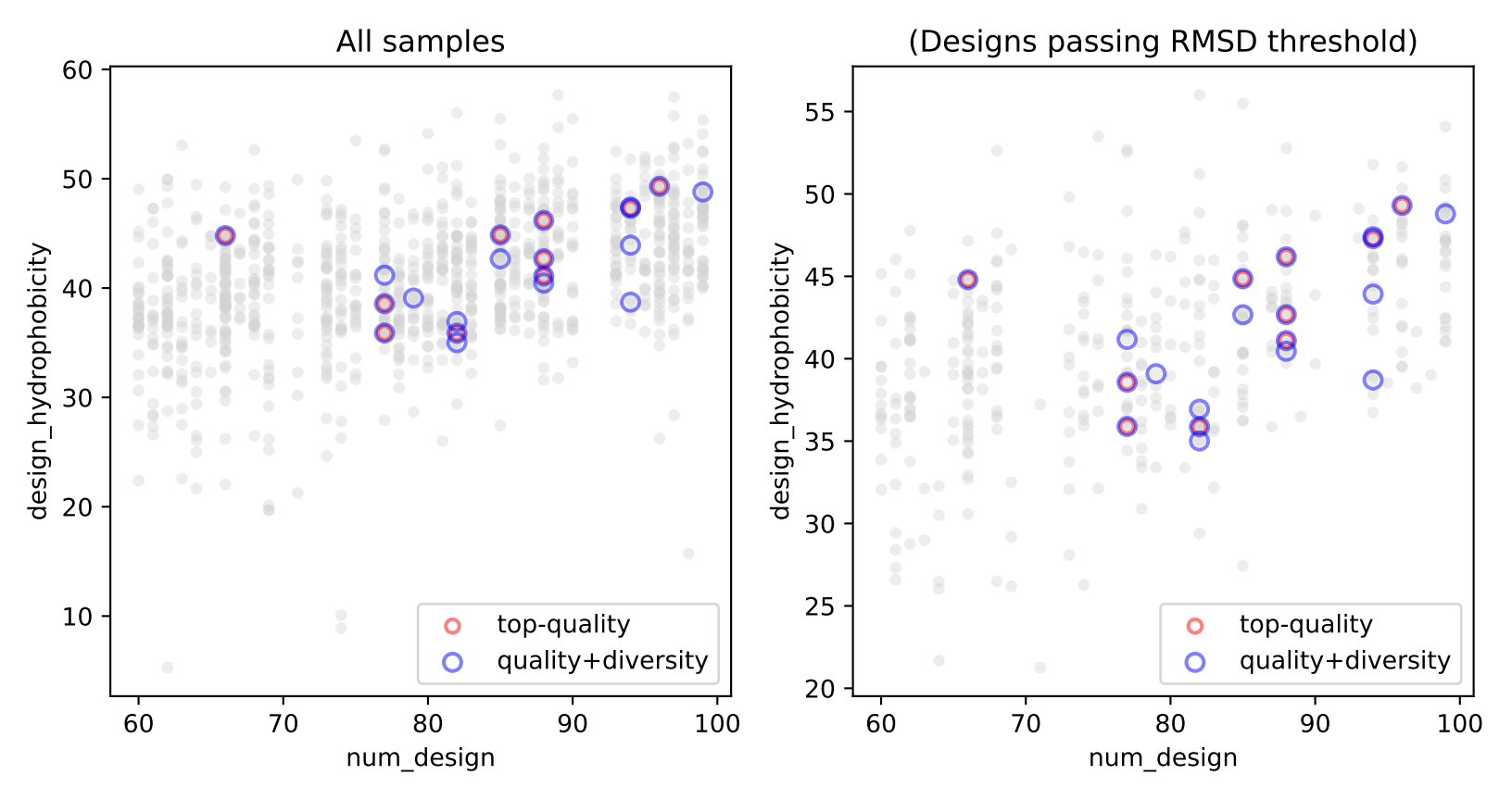

All samples
top-quality
quality+diversity
min_design_to_target_pae
num_design
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
60
70
80
90
100
(Designs passing RMSD threshold)
top-quality
quality+diversity
min_design_to_target_pae
num_design
12
10
8
6
4
2
60
70
80
90
100


All samples
(Designs passing RMSD threshold)
design_to_target_iptm
num_design
top-quality
quality+diversity
0.7
0.6
0.5
0.4
0.3
0.2
0.1
60
70
80
90
100


All samples
(Designs passing RMSD threshold)
design_iptm
num_design
top-quality
quality+diversity
0.7
0.6
0.5
0.4
0.3
0.2
0.1
60
70
80
90
100


All samples
(Designs passing RMSD threshold)
design_iiptm
num_design
top-quality
quality+diversity
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
60
70
80
90
100

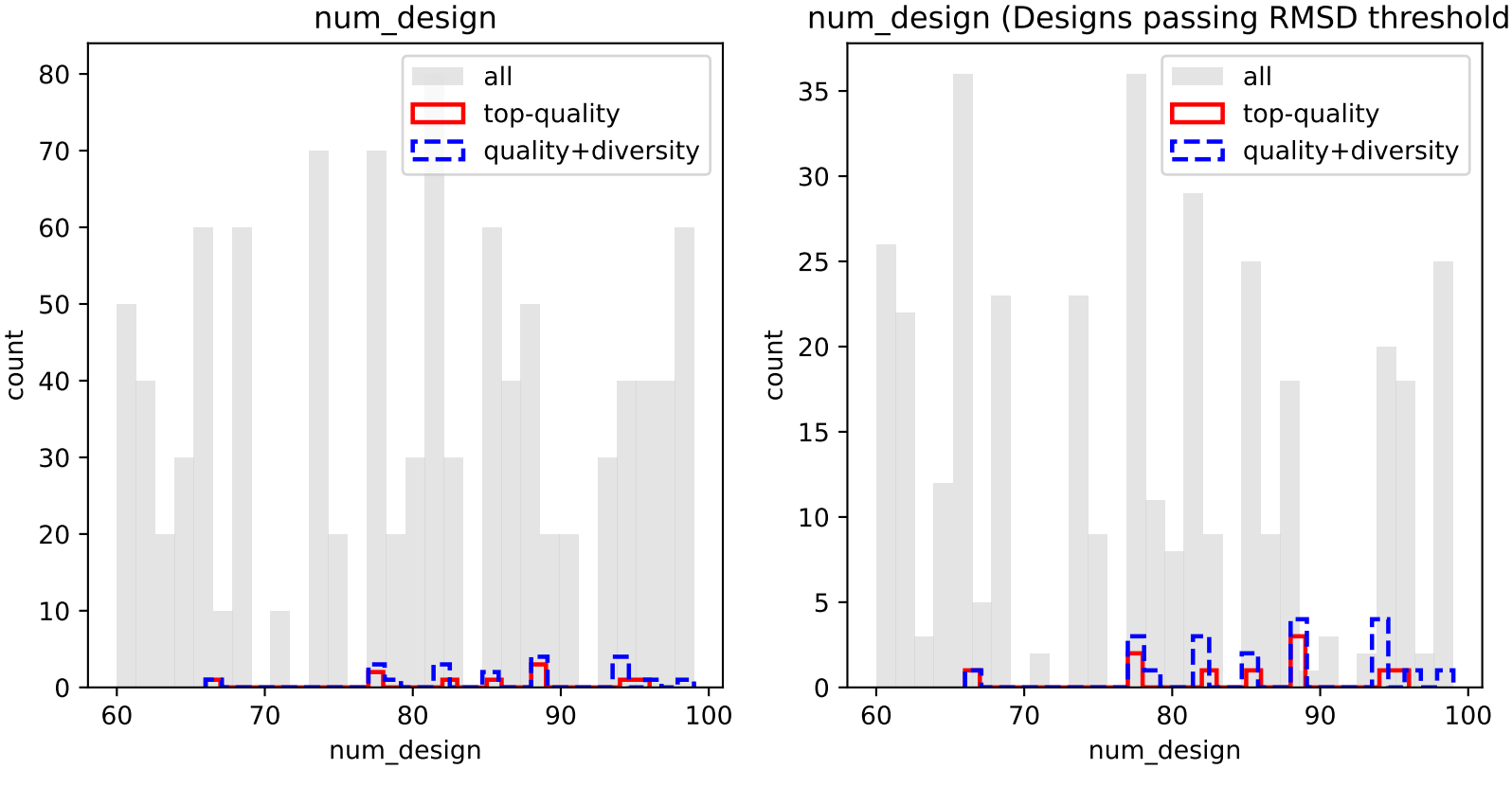

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_largest_hydrophobic_patch_refolded
num_design
2500
2000
1500
1000
500
60
70
80
90
100

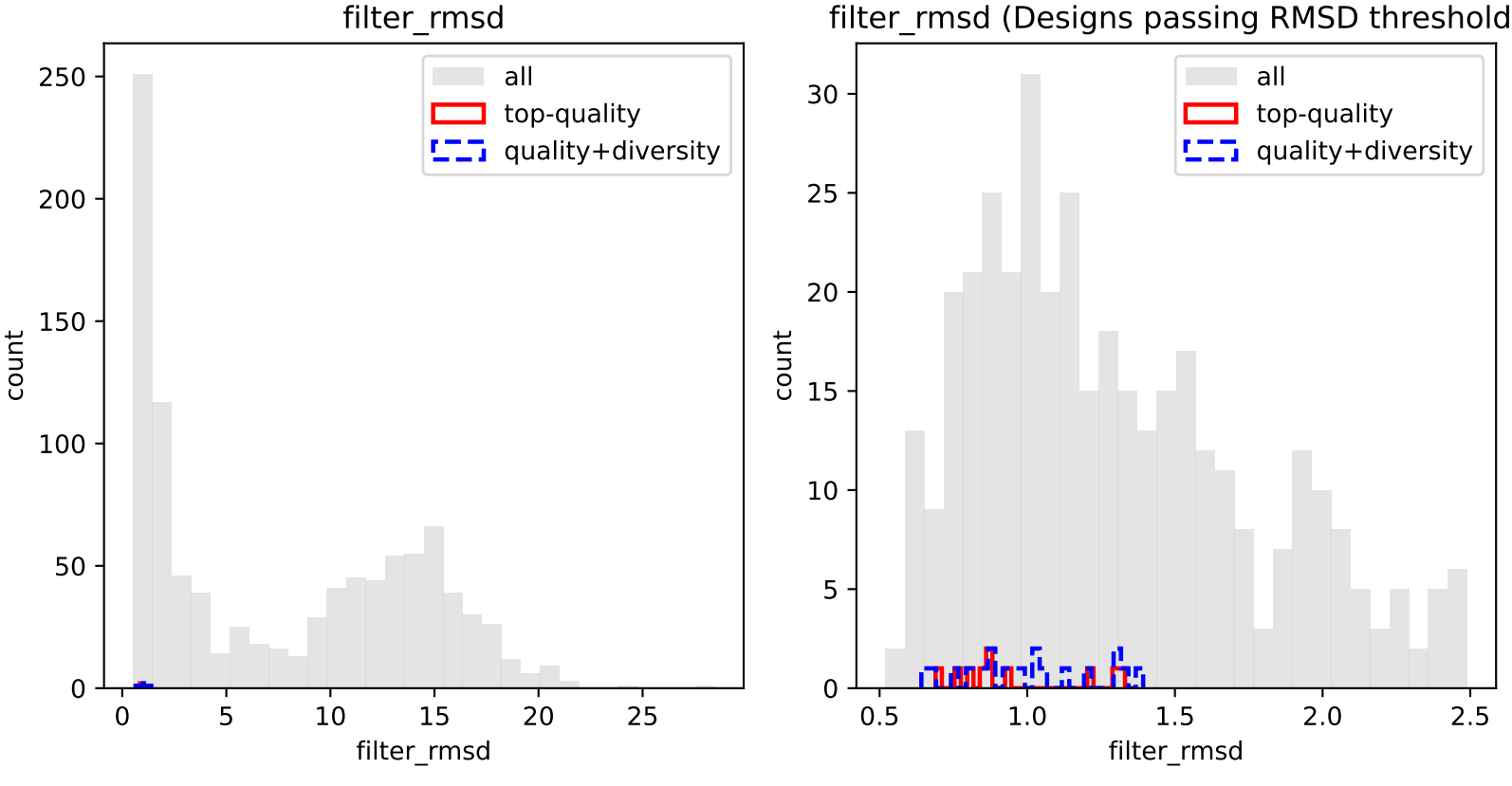

All samples
(Designs passing RMSD threshold)
design_hydrophobicity
num_design
top-quality
quality+diversity
60
70
80
90
100
10
20
30
40
50
55
25
35
45

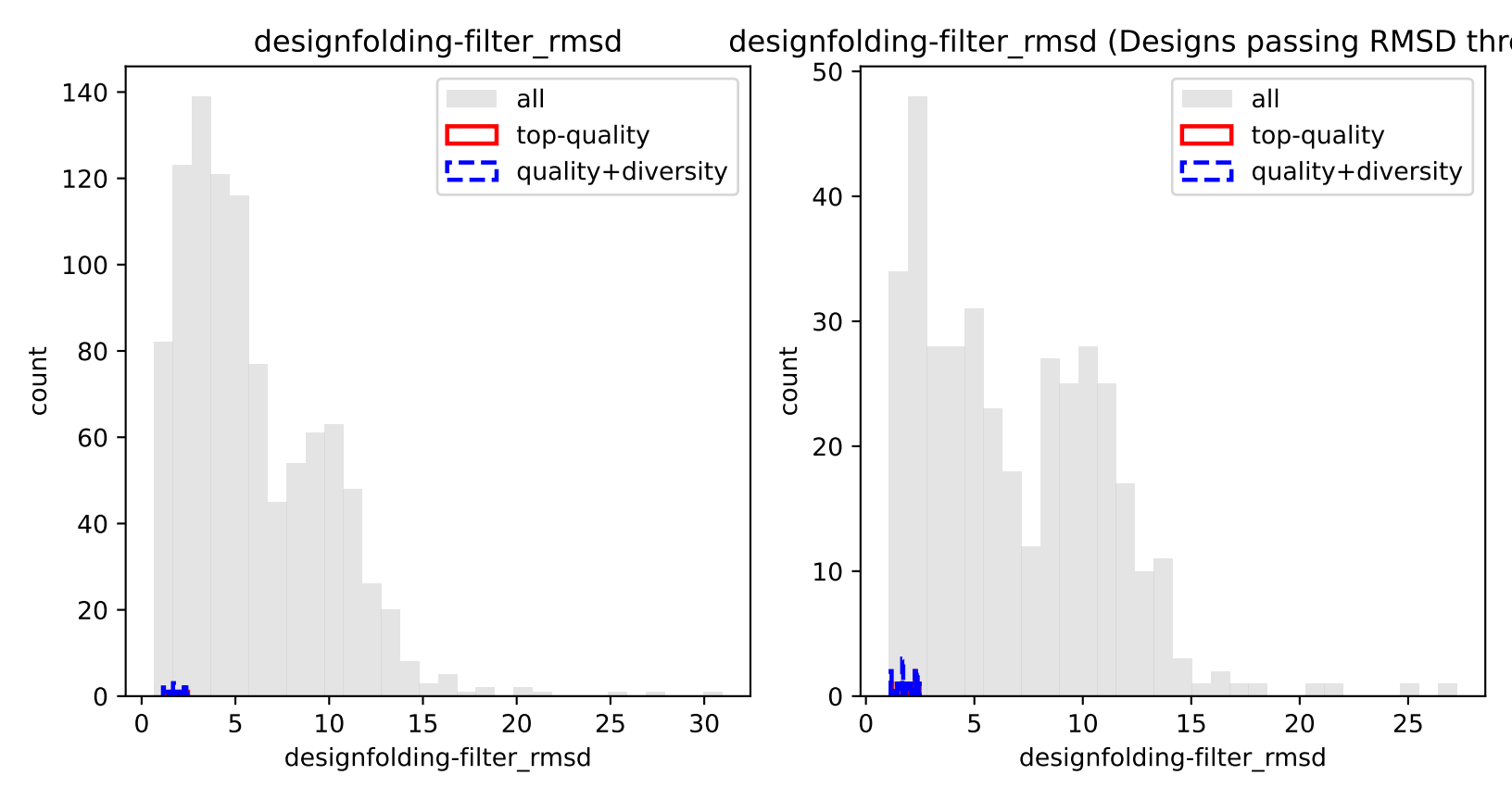
Metric Distributions

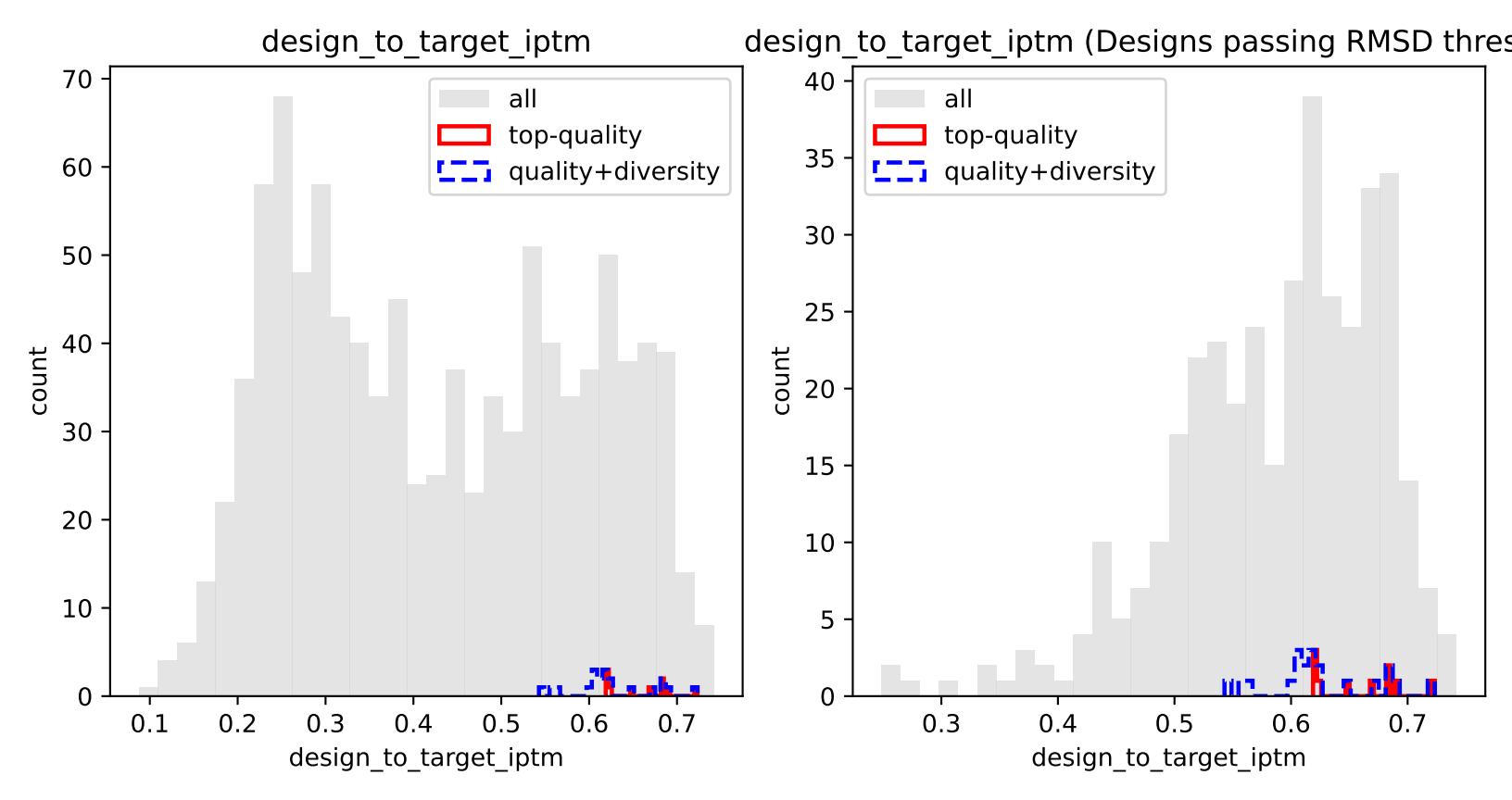
Histograms

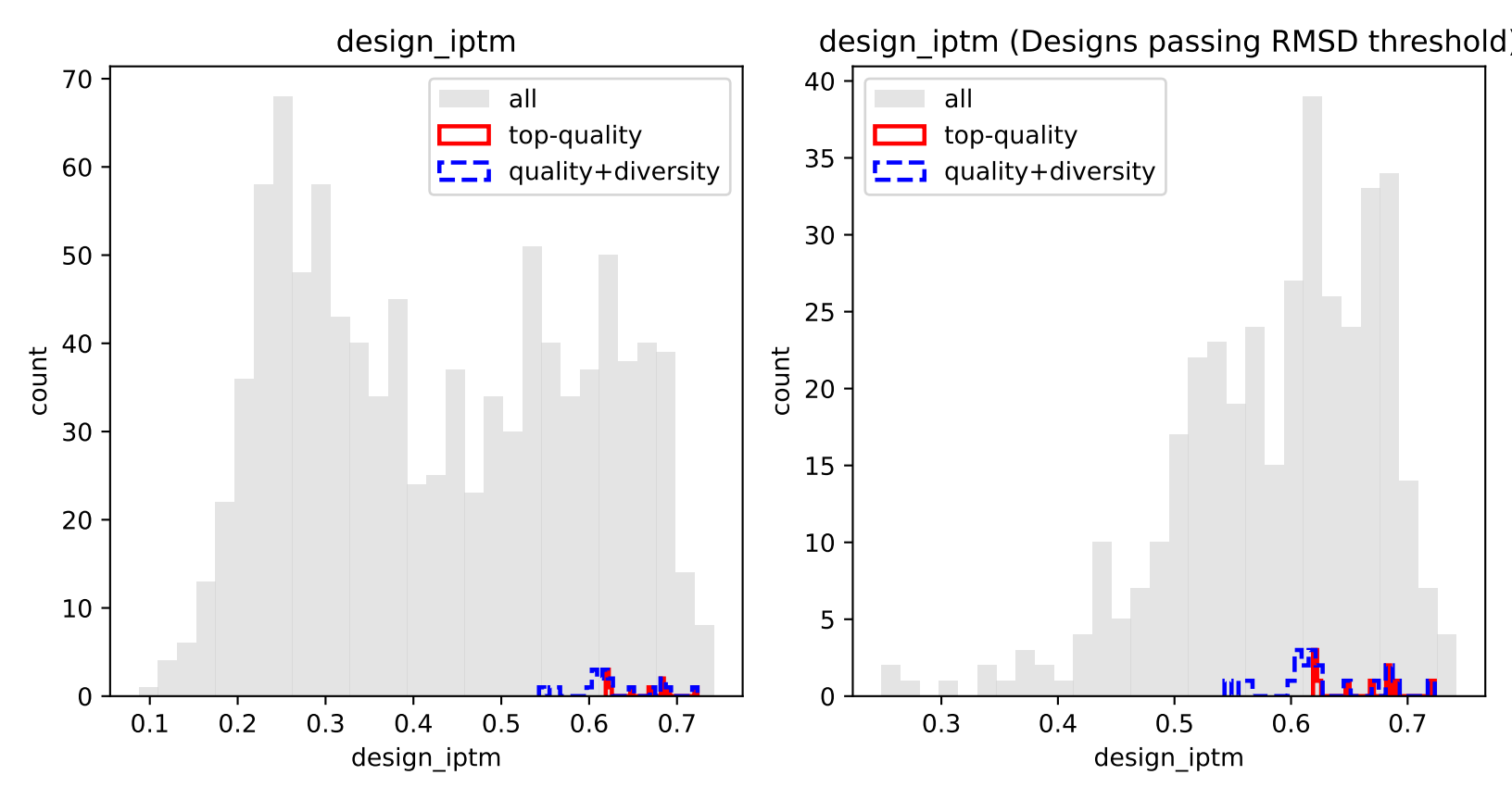
Distribution of each metric across all designs (grey) with overlays for Top-quality (red outline) and Diversity-optimised (blue dashed) subsets. The right panel repeats the histogram but only for designs that pass the RMSD threshold.

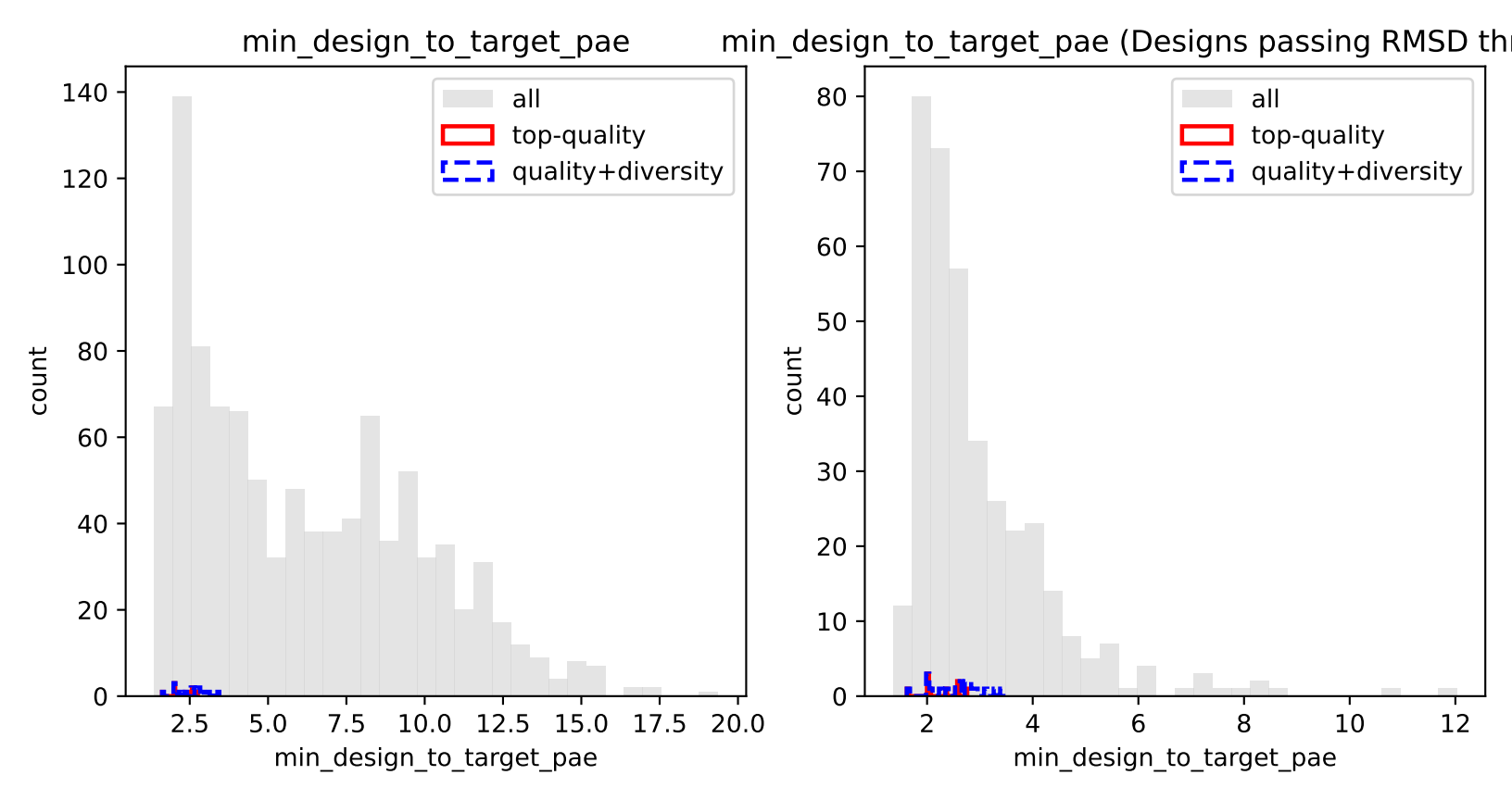

num_design
all
top-quality
quality+diversity
count
num_design
num_design (Designs passing RMSD threshold)
all
top-quality
quality+diversity
count
num_design

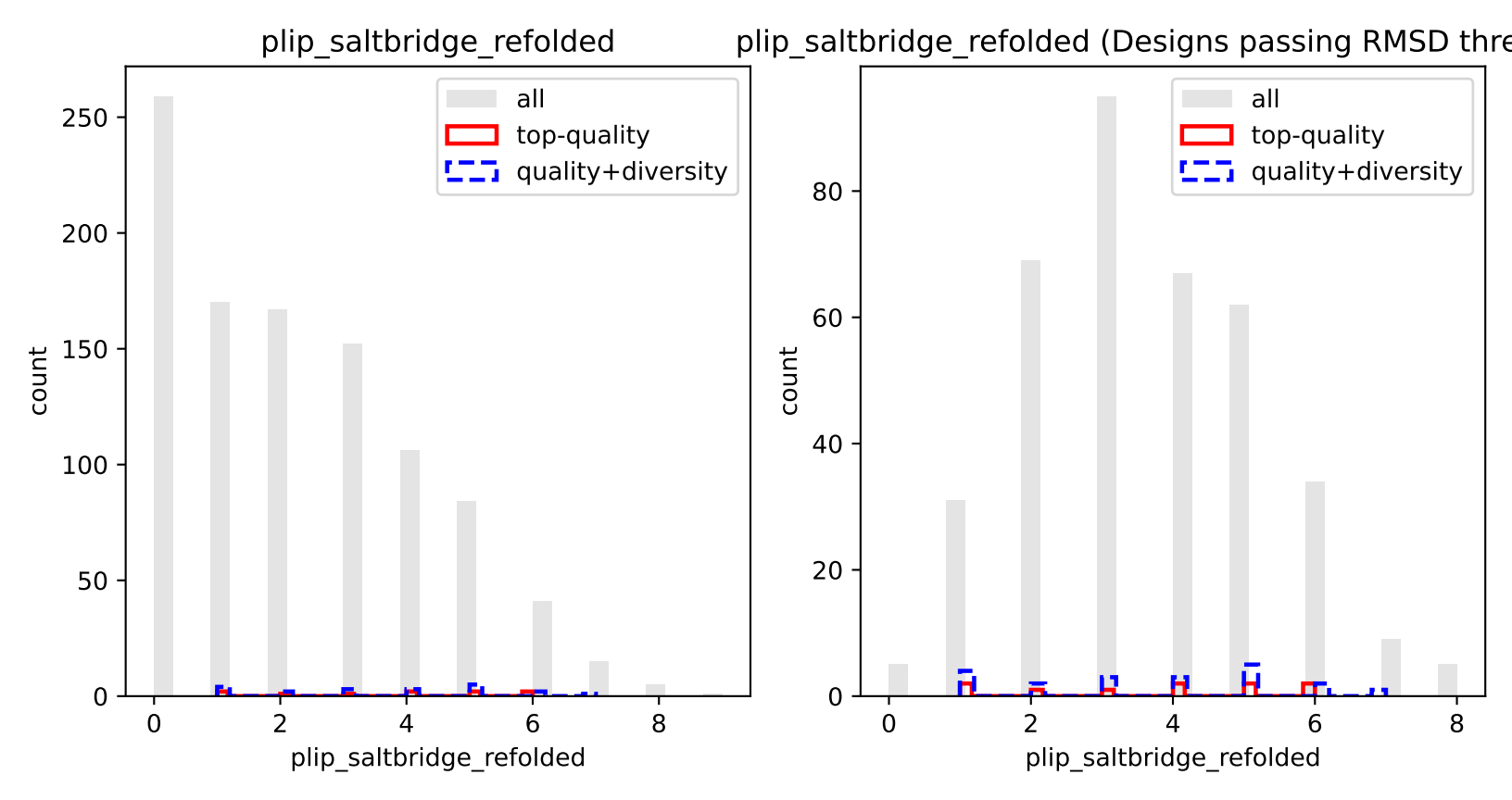

filter_rmsd
all
top-quality
quality+diversity
count
filter_rmsd
filter_rmsd (Designs passing RMSD threshold)
all
top-quality
quality+diversity
count
filter_rmsd

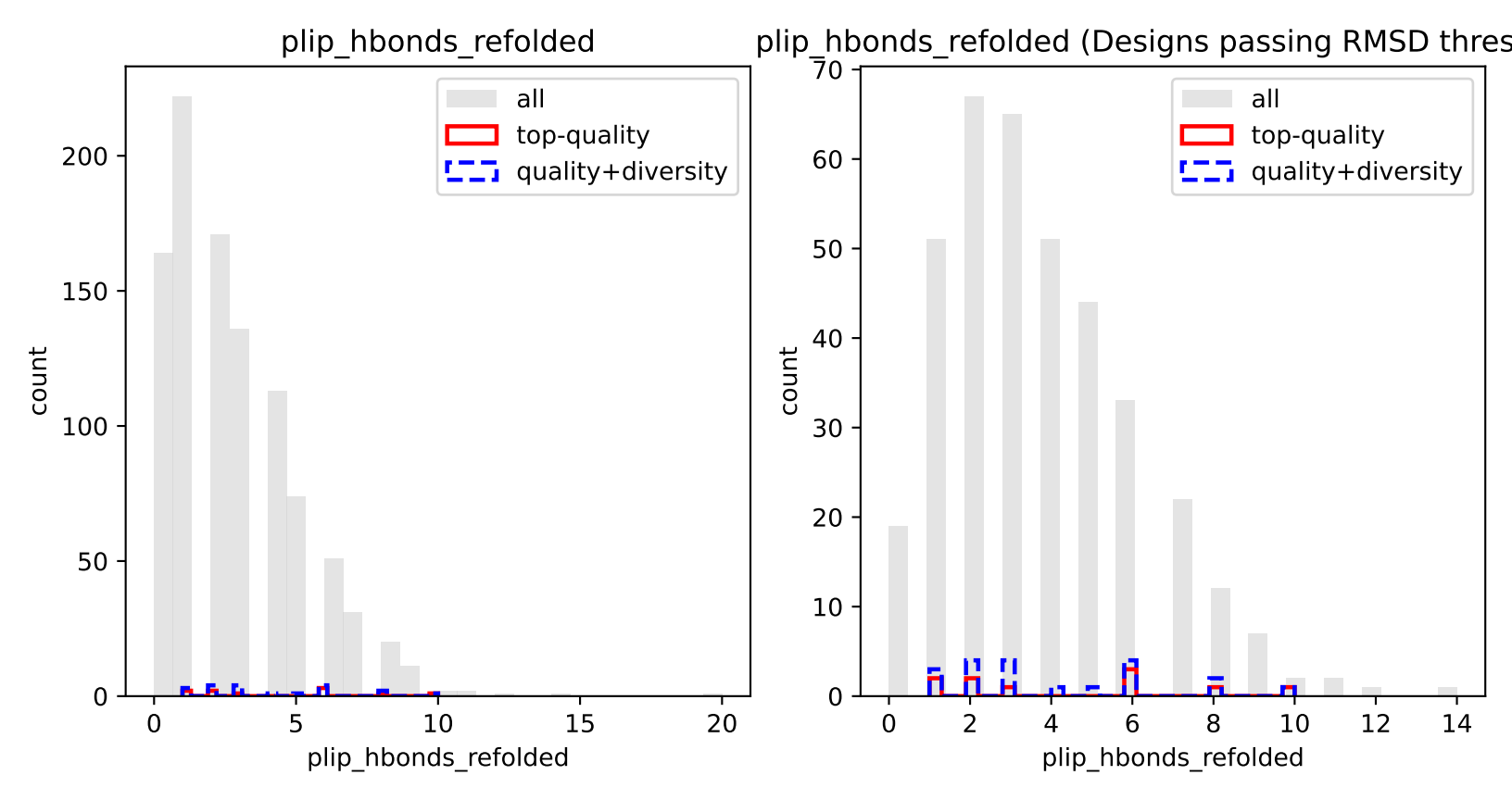

designfolding-filter_rmsd
all
top-quality
quality+diversity
count
designfolding-filter_rmsd
designfolding-filter_rmsd (Designs passing RMSD thr
all
top-quality
quality+diversity
count
designfolding-filter_rmsd

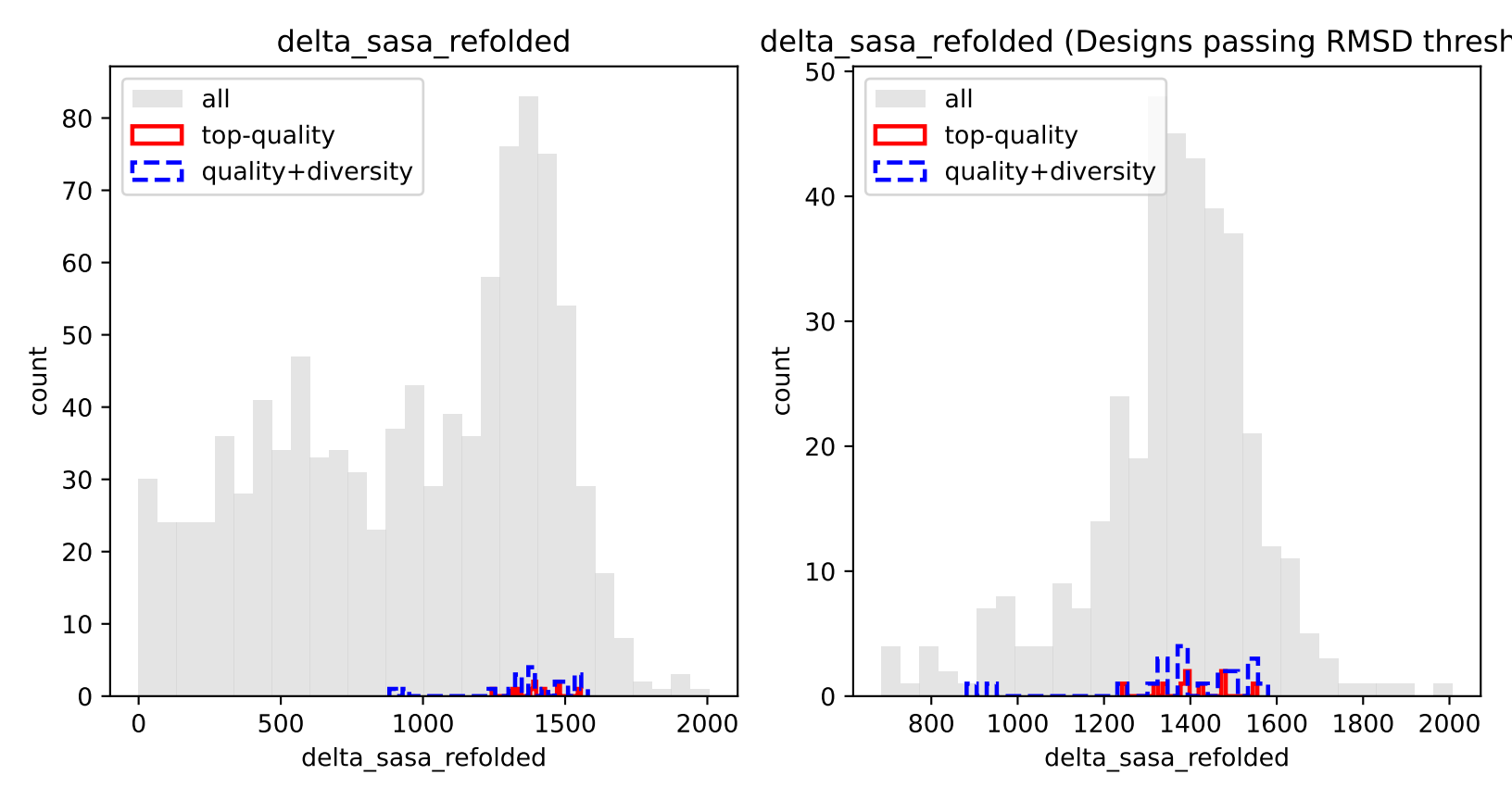

design_to_target_iptm
all
top-quality
quality+diversity
count
design_to_target_iptm
design_to_target_iptm (Designs passing RMSD thres
all
top-quality
quality+diversity
count
design_to_target_iptm

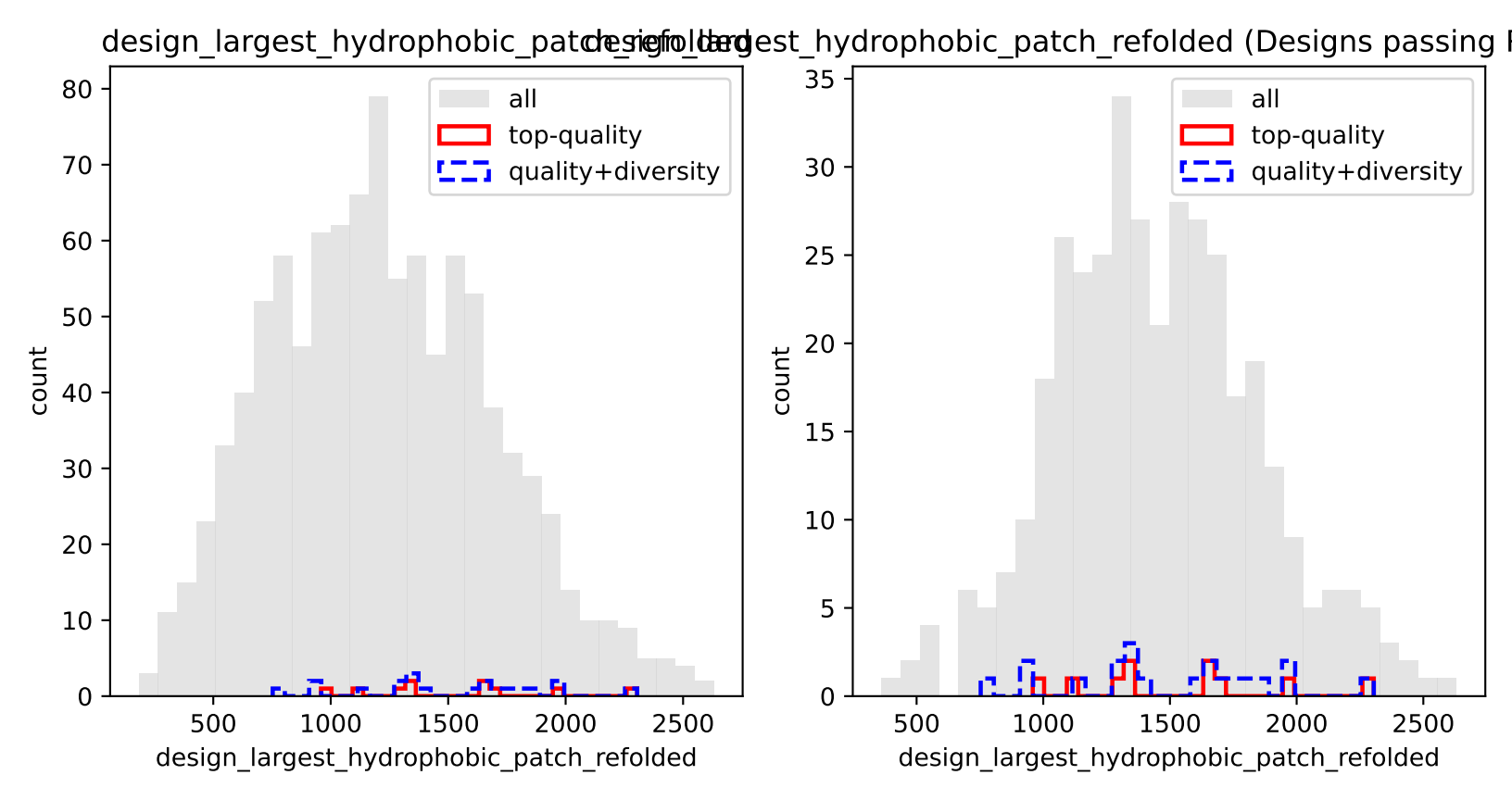

design_iptm
all
top-quality
quality+diversity
count
design_iptm
design_iptm (Designs passing RMSD threshold)
all
top-quality
quality+diversity
count
design_iptm

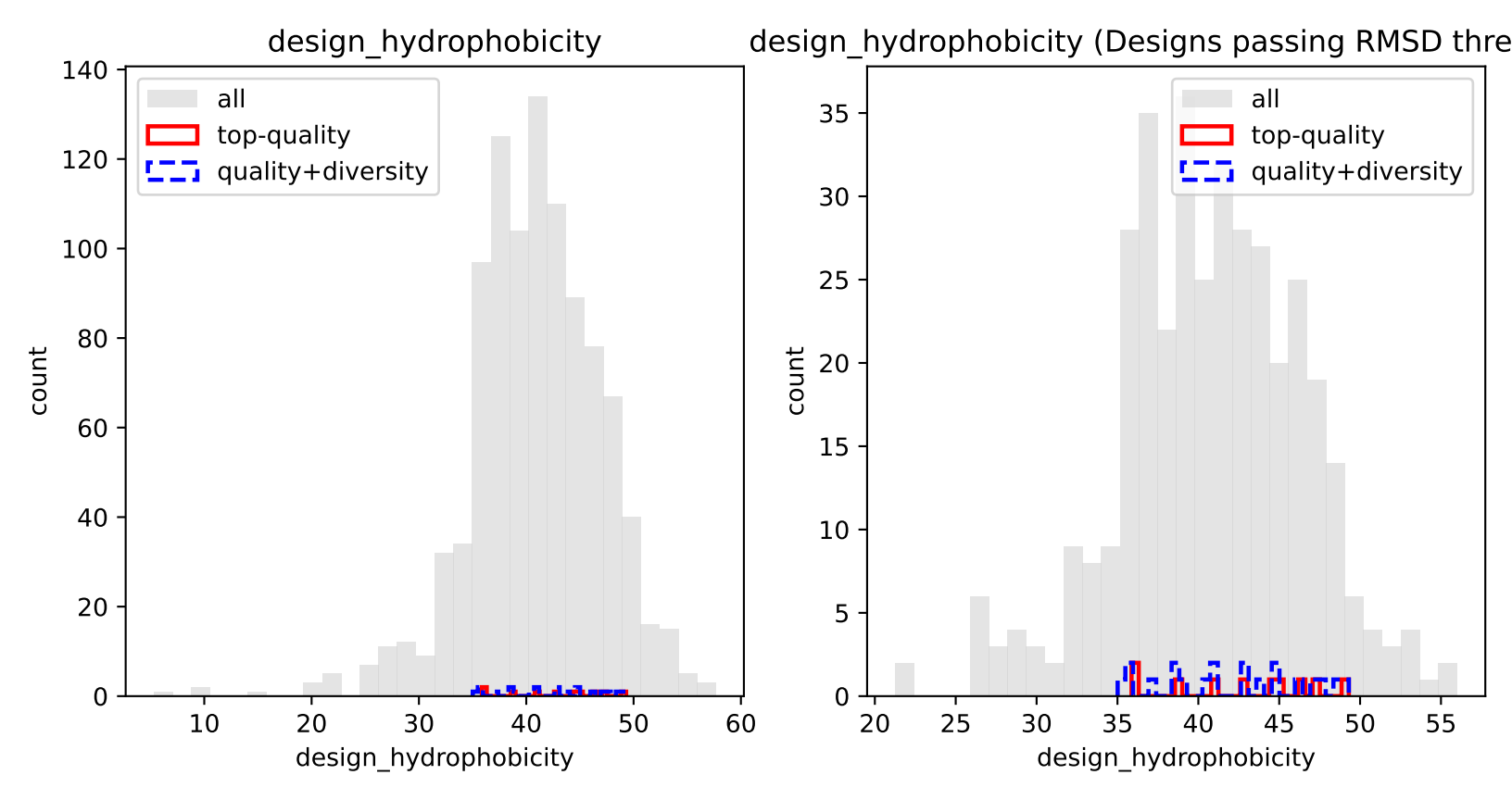

min_design_to_target_pae
min_design_to_target_pae (Designs passing RMSD thr
all
top-quality
quality+diversity
count
min_design_to_target_pae
0
20
40
60
80
100
120
140
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10
30
50
70
2
4
6
8
10
12


plip_saltbridge_refolded
plip_saltbridge_refolded (Designs passing RMSD thre
all
top-quality
quality+diversity
count
plip_saltbridge_refolded
0
50
100
150
200
250
20
40
60
80
2
4
6
8


plip_hbonds_refolded
all
top-quality
quality+diversity
count
plip_hbonds_refolded
plip_hbonds_refolded (Designs passing RMSD thres
all
top-quality
quality+diversity
count
plip_hbonds_refolded


delta_sasa_refolded
all
top-quality
quality+diversity
count
delta_sasa_refolded
0
500
1000
1500
2000
0
10
20
30
40
50
60
70
80
delta_sasa_refolded (Designs passing RMSD thresh
all
top-quality
quality+diversity
count
delta_sasa_refolded
800
1000
1200
1400
1600
1800
2000
0
10
20
30
40
50


design_largest_hydrophobic_patch_refolded
design_largest_hydrophobic_patch_refolded (Designs passing F
all
top-quality
quality+diversity
count
design_largest_hydrophobic_patch_refolded
0
10
20
30
40
50
60
70
80
500
1000
1500
2000
2500
0
5
10
15
20
25
30
35


design_hydrophobicity
all
top-quality
quality+diversity
count
design_hydrophobicity
design_hydrophobicity (Designs passing RMSD thre
all
top-quality
quality+diversity
count
design_hydrophobicity